Network Science

Lecture 05: Social Context and Link Formation

Prof. Dr. Michael Granitzer

Dr. Kanishka Ghosh Dastidar

Dr. Jelena Mitrovic

From Structure to Context

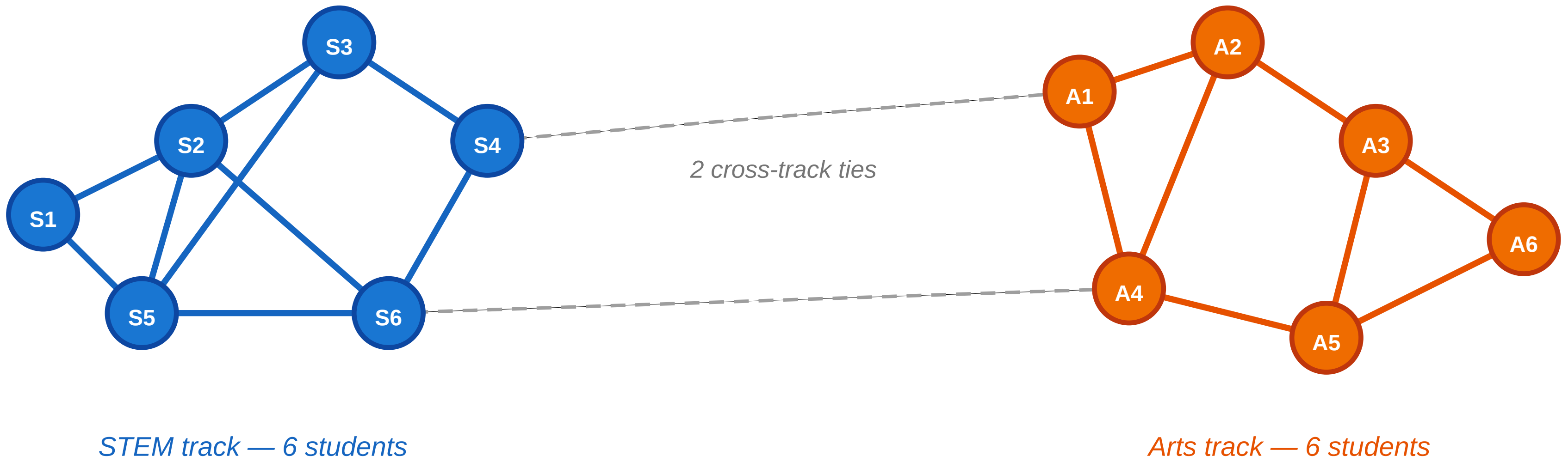
In Lectures 03 and 04 we classified edges (strong vs. weak) and groups (communities). The graph was colorless — every node was interchangeable. Today we ask: what happens when nodes carry *attributes* and edges carry *reasons*?

Consider a small classroom.

A Motivating Scenario

A small classroom: friendships and attributes

circles = students · color = track (blue = STEM, orange = Arts) · lines = friendships



Questions we'd like to answer from this picture

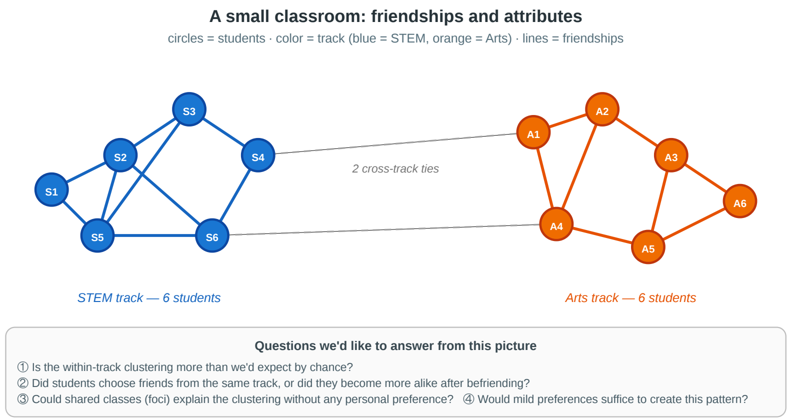
① Is the within-track clustering more than we'd expect by chance?
② Did students choose friends from the same track, or did they become more alike after befriending?
③ Could shared classes (foci) explain the clustering without any personal preference? ④ Would mild preferences suffice to create this pattern?

Twelve students, split equally between STEM (blue) and Arts (orange). Most friendships are within-track; only two cross-track ties exist. The clustering is visible — but why is it there?

Four Questions from One Picture

1. **Measurement.** Is the within-track clustering more than we would expect by chance — or could it happen under random mixing?
2. **Mechanism.** Did students choose friends from the same track (**selection**), or did friends influence each other to pick the same track (**socialization**)?
3. **Context.** Could shared classes or labs (**foci**) explain the clustering without any personal preference for similarity?
4. **Dynamics.** If every student has only a mild preference for same-track friends, could that suffice to produce this sharp segregation?

Every concept in today's lecture answers one of these.



A Different Kind of Gap

In L03–L04 the gap between theory and measurement was **computational**: MaxSTC and modularity maximization are NP-hard, so we used polynomial-time heuristics.

In L05 the gap is **causal**: even with infinite compute, a cross-sectional snapshot cannot tell us *why* a tie formed. The fix is not a better algorithm — it is richer data.

Question	What we want to know	What a snapshot shows	What we need
Homophily?	Is tie formation biased toward similarity?	Within-group edge fraction	Random-mixing baseline
Selection or socialization?	Which came first — similarity or the tie?	Both present simultaneously	Longitudinal data
Foci or preference?	Did shared context cause the tie?	Shared memberships and ties	Affiliation network model
Local → global?	Can mild preferences produce sharp segregation?	A segregated pattern	A generative model (Schelling)

Takeaway

L03–L04 asked "what structure is there?" and hit a computational wall. L05 asks "why is the structure there?" and hits a causal wall. Network analysis requires both algorithmic tools and causal reasoning about the mechanisms behind the observed graph.

Homophily

> **Guiding Question:** Do people form ties because they are similar — and how would we know?

Homophily — Formal Definition

Homophily. Edge formation is biased toward similarity in a node attribute x. Formally, for a categorical attribute with class shares $p_1, \ldots, p_k$:

- **Random-mixing baseline:** the probability that a random tie connects same-class nodes is

$$H_{\text{base}} = \sum_{i=1}^{k} p_i^2$$

- **Observed homophily:** the fraction of actual ties that connect same-class nodes, H_{obs}.
- **Homophily index:** how much the observed same-class rate exceeds the random baseline, normalized:

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$$r = \frac{H_{\text{obs}} - H_{\text{base}}}{1 - H_{\text{base}}}$$

$r = 0$: random mixing. $r = 1$: perfect segregation. $r < 0$: heterophily (preference for *different* attributes).

Homophily Baseline Trap

The same observed within-group share can mean very different things depending on the population composition.

Population shares	Random baseline H_{base}	If $H_{\text{obs}} = 0.85$	Interpretation
50% / 50%	$0.5^2 + 0.5^2 = 0.50$	$r = 0.70$	Strong excess homophily
90% / 10%	$0.9^2 + 0.1^2 = 0.82$	$r = 0.17$	Only mild excess homophily

Never interpret "85% of ties are within-group" without asking: **85% compared to what baseline?**

Visual Intuition

Homophily as similarity-biased tie formation

Solid edges are frequent within-group ties; dashed edges are rarer cross-group ties.

Within-group edges (colored) are far more frequent than cross-group edges (gray). But frequency alone is insufficient — we need the random-mixing baseline to decide whether this exceeds chance.

Back to the Classroom

In the scenario, both tracks have 6 students ($p_{\text{STEM}} = p_{\text{Arts}} = 0.5$).

- $H_{\text{base}} = 0.5^2 + 0.5^2 = 0.50$
- The figure has 17 within-track edges out of 19 total: $H_{\text{obs}} = 17/19 \approx 0.89$
- $r = (0.89 - 0.50)/(1 - 0.50) = 0.78$

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The homophily index $r = 0.78$ is strong — far above random mixing. But it does not tell us *why*. Is it selection, socialization, or shared context?

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Quick Check

In a school of 200 students, 60% are in track A and 40% in track B. Of all friendships, 78% are within-track.

1. Compute the random-mixing baseline H_{base}.
2. Compute the homophily index r.
3. Is r strong evidence of homophily? What could make it misleading?

Quick Check — Solution

$$H_{\text{base}} = 0.6^2 + 0.4^2 = 0.36 + 0.16 = 0.52$$

$$r = \frac{0.78 - 0.52}{1 - 0.52} = \frac{0.26}{0.48} \approx 0.54$$

$r = 0.54$ is moderate-to-strong evidence of homophily. It could be misleading if:

- students were to tracks based on pre-existing friend groups (reverse causation)

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3. $r = 0.54$ is moderate-to-strong evidence of homophily. It could be misleading if:

- the two tracks occupy different buildings (shared context, not personal preference)
- students were *assigned* to tracks based on pre-existing friend groups (reverse causation)
- a third factor (e.g. family background) causes both track choice and friendship patterns (confounding)

Takeaway

Homophily is not just a visual pattern — it is a statistical claim that tie formation is biased toward similarity *relative to a random-mixing baseline*. The homophily index r quantifies the excess, but it cannot identify the mechanism.

Study: Echo Chambers as Homophily

Source: Cinelli et al. 2021; Nyhan et al. 2023

Question	Do social platforms expose users mostly to like-minded peers and sources?
Evidence A	PNAS cross-platform study: Facebook, Twitter, Reddit, Gab; >100M pieces of content
Network signal	Homophily: users interact with peers holding similar topical or political leanings
Diffusion signal	Information tends to circulate toward like-minded users rather than crossing groups

Outcome: Echo chambers are visible as dense, homophilic interaction regions. Facebook and Twitter show strong clustering by user leaning; Reddit is less segregated in direct comparison.

Evidence B	Nature field experiment on Facebook during the 2020 U.S. election
Scale	23,377 consenting users; exposure to like-minded sources reduced by about one-third
Result	Cross-cutting exposure increased, but attitudes and false-belief measures did not measurably change

Interpretation: Echo chambers are a real structural exposure pattern, but exposure alone is not a simple causal switch for polarization.

Caution: The network concept is clearer than the causal claim. Dense homophily can reinforce information, but changing feed exposure for a few weeks may be too weak, too late, or too indirect to change stable political attitudes.

Selection and Socialization

> **Guiding Question:** How can we distinguish whether similarity caused the tie or the tie caused the similarity?

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$$\text{homophily} = \text{selection} + \text{socialization} + \text{context}$$

Three Mechanisms, One Pattern

Selection. People choose to form ties with others who are *already* similar to them. Similarity precedes the tie.

Socialization (social influence). People become *more similar* to their contacts after the tie forms. The tie precedes similarity.

Contextual correlation. A shared environment (school, workplace, neighborhood) independently causes both the attribute and the tie. Neither causes the other directly.

Same Snapshot, Three Stories

Suppose we observe: **STEM students mostly befriend STEM students.**

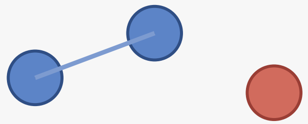
Mechanism	Concrete story	What data would help?
Selection	Students first choose STEM, then choose friends from the same track.	Friendships observed *after* track choice
Socialization	Friends influence each other to choose the same elective or track.	Track changes observed *after* friendships
Contextual correlation	STEM students share labs, schedules, and buildings, so they meet more often.	Timetables, room assignments, shared courses

The graph snapshot is identical in all three stories. The difference is the **time order** and the **unobserved context**.

Causal Diagrams

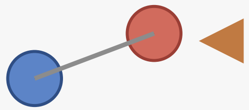
Why observed similarity can arise in different ways

Selection



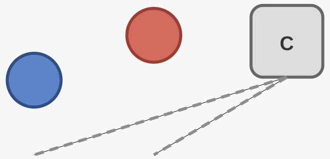
similar actors choose each other

Socialization



connected actors become more alike

Context



a third factor affects both nodes

Empirical Test: Christakis & Fowler — Obesity in Social Networks

Christakis & Fowler (2007), *The Spread of Obesity in a Large Social Network Over 32 Years*, New England Journal of Medicine 357, 370–379.

They analyzed the **Framingham Heart Study** social network — 12 067 people tracked over **32 years** with repeated measurements of both friendship ties and body mass index. This made it possible to observe the *temporal sequence* of tie formation and weight change.

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Headline result. A person's probability of becoming obese increased by **57%** if a friend became obese in the same interval. The effect was stronger for mutual friends and showed **directionality** — it operated in the direction of the friendship nomination, suggesting socialization (influence) rather than pure selection.

What the Study Could Not See

1. Confounders. Cohen-Cole & Fletcher (2008) and Lyons (2011) showed that the same statistical method would "detect" social contagion of **height** and **acne** — traits that clearly do not spread through social influence. The method may confuse *shared environment* with *social influence* when the confounders evolve over time.

2. Homophily in tie formation. The study adjusted for *existing* attribute similarity but could not fully exclude the possibility that people selectively *formed* or *maintained* ties with others on similar weight trajectories — a dynamic form of selection invisible to static controls.

3. Missing network. The Framingham data records nominated friends, not the full social network. Influence may flow through unobserved ties (coworkers, family, neighbors) that correlate with both friendship nominations and weight change.

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Quick Check

You observe that smokers tend to befriend other smokers in a college dorm.

1. What kind of data would you need beyond a single snapshot to distinguish selection from socialization?
2. Sketch one observation pattern that would favor *selection* and one that would favor *socialization*.
3. Even with longitudinal data, what kind of confound would remain hard to rule out?

Quick Check — Solution

Longitudinal observations — friendships and smoking status at multiple time points, with newly formed ties and newly adopted behavior visible.

- : New friendships disproportionately form between people who already share smoking status.
- : People who befriend smokers become more likely to start smoking in the next observation period.

3. Shared — e.g. dorm floor placement causing both friendship and smoking exposure simultaneously. Even with temporal data, confounders that evolve over time alongside both the network and the attribute are hard to rule out without randomization.

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3. Shared *environment* — e.g. dorm floor placement causing both friendship and smoking exposure simultaneously. Even with temporal data, confounders that evolve over time alongside both the network and the attribute are hard to rule out without randomization.

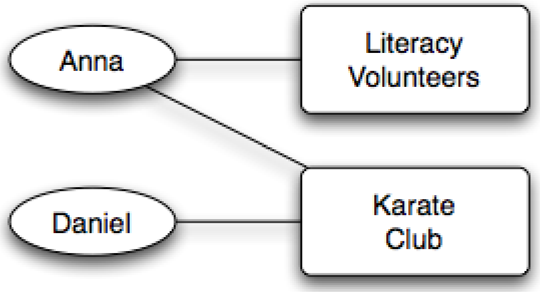
Takeaway

Cross-sectional data cannot distinguish selection from socialization — the same homophily index r is consistent with both. Longitudinal network data is far more informative, but even multi-decade panel studies cannot fully eliminate confounding by evolving shared environments.

Affiliation Networks

> **Guiding Question:** How do we represent the social settings — classes, clubs, workplaces — in which ties form?

Bipartite Affiliation Model



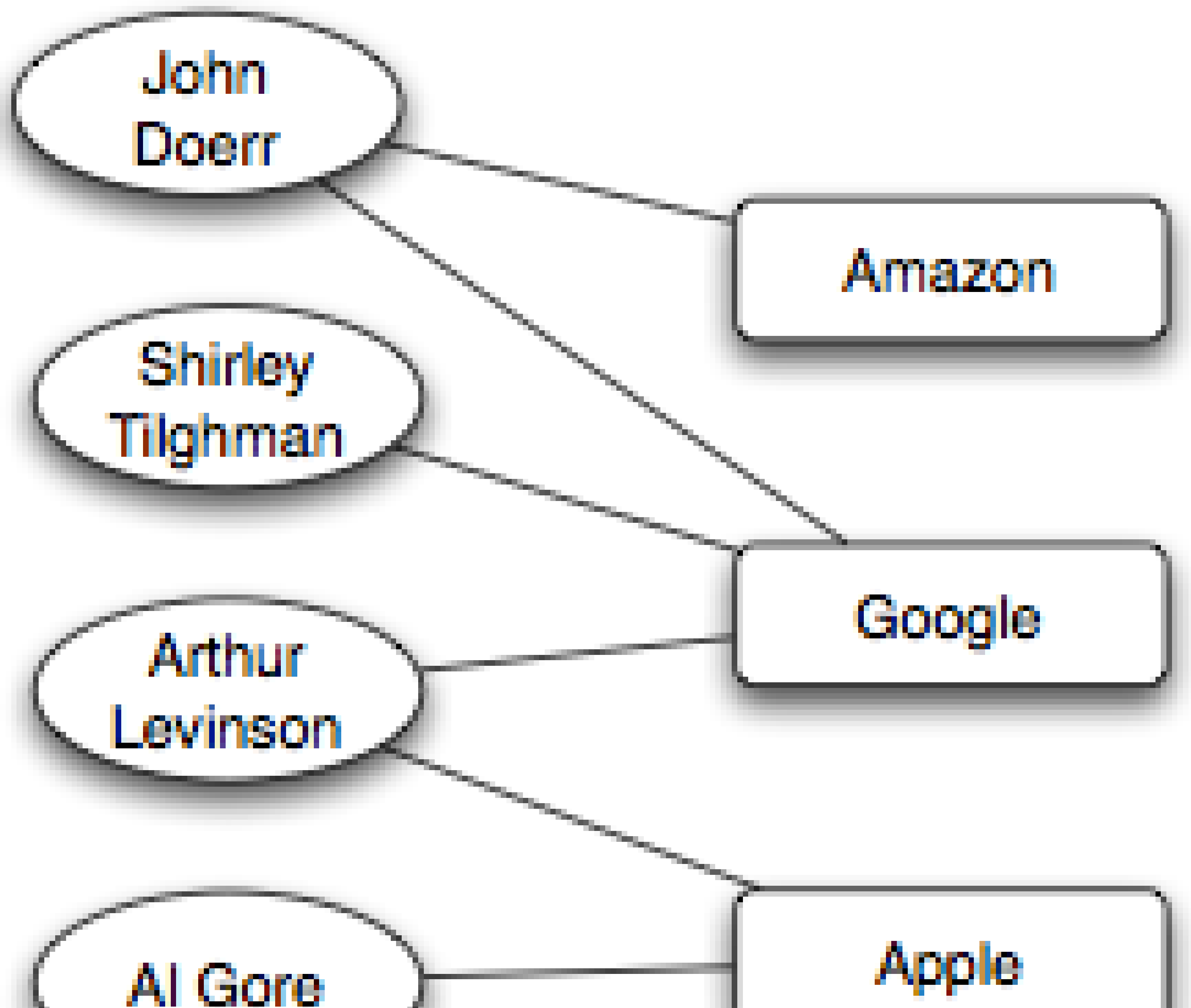
Source: Easley & Kleinberg 2010

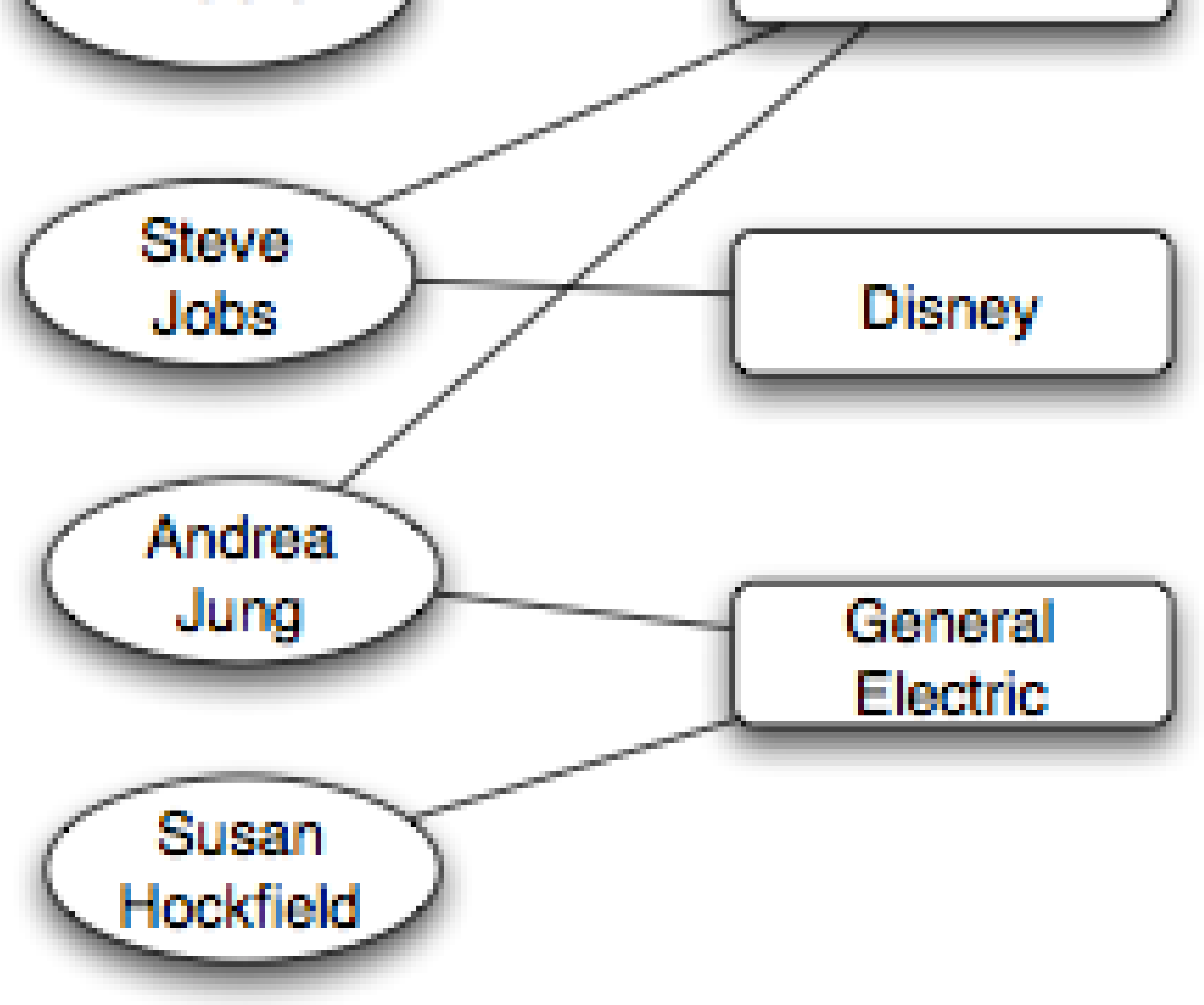
An **affiliation network** is a bipartite graph $G = (P \cup F, E)$:

- P: actor nodes (persons)
- F: focus nodes (groups, locations, classes, organizations)
- E: edges only between P and F – "person p participates in focus f"

Two persons sharing a focus creates a *potential* for a person–person tie.

Example — Corporate Board Interlocks


John Doerr
Amazon
Shirley Tilghman
Google
Arthur Levinson
Al Gore
Apple



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Reading the Board Network

An affiliation network can be projected in two different ways.

Projection	Nodes	Edge means	Example question
Director network	People	Two directors sit on the same board	Who can exchange career information or governance norms?
Company network	Firms	Two firms share at least one director	Where can strategy, norms, or practices diffuse?

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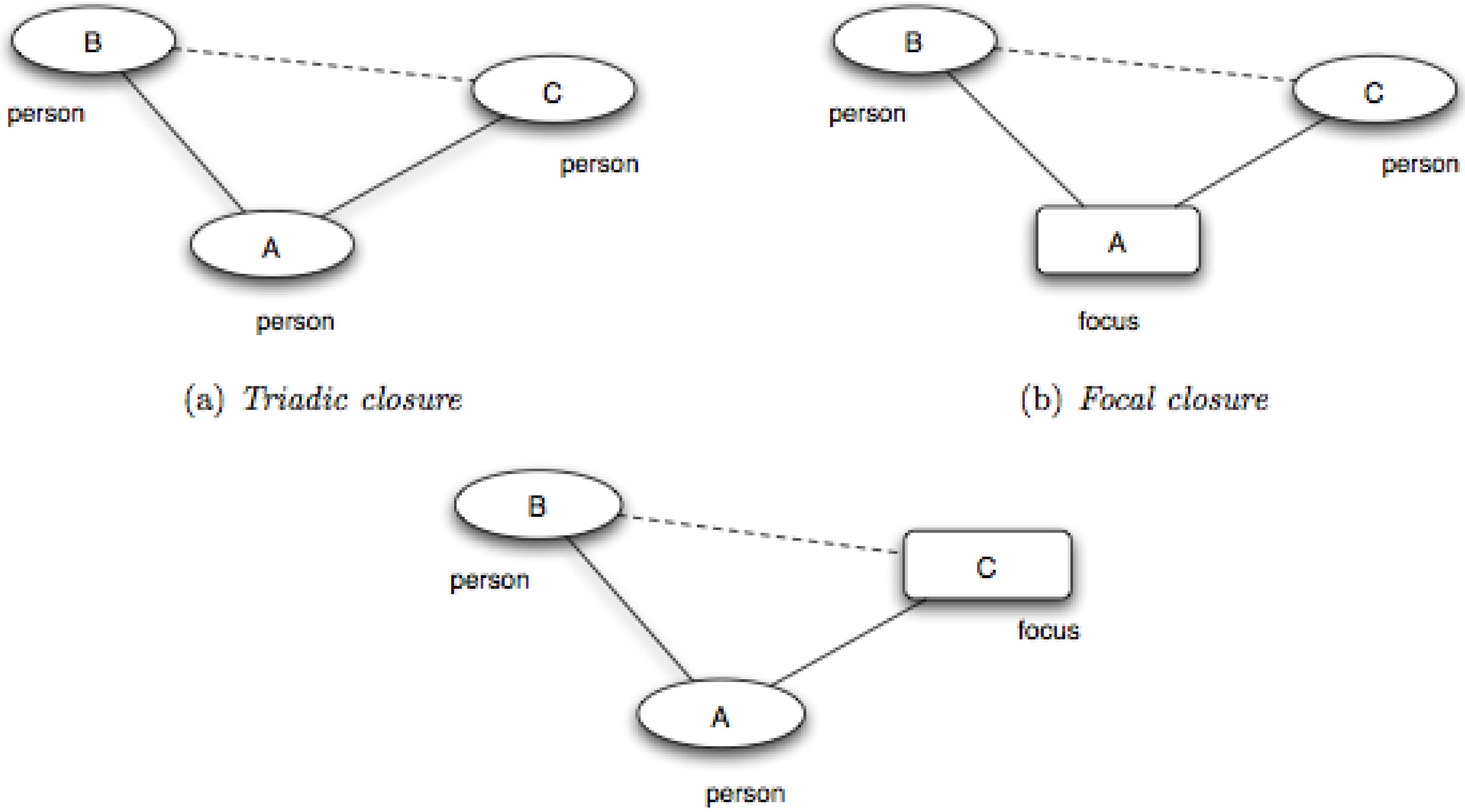
Important: an edge in the projection is not automatically a friendship or causal influence. It is an **opportunity for exposure** created by shared context.

Three Closure Processes



Source: Easley & Kleinberg 2010

Triadic closure. Friend-of-friend becomes friend — a person–person–person path closes into a triangle. *(Same mechanism as Lecture 03.)*

Focal closure. Two persons sharing a focus become friends — a person–focus–person path closes into a person–person tie. *(New — the focus mediates the introduction.)*

Membership closure. A friend introduces you to a focus you did not belong to — a person–person–focus path closes into a person–focus tie. *(New — the friend mediates membership.)*

Closure Processes — Classroom Examples

Closure type	Open path	What closes?	Classroom example
Triadic	student → friend → student	A new friendship	"My lab partner introduces me to their friend."
Focal	student → elective → student	A new friendship	"We sit in the same statistics elective and start talking."
Membership	student → friend → club	A new membership	"My friend invites me to join the robotics club."

Only triadic closure needs a shared friend. Focal closure can create ties among strangers through repeated exposure in the same setting.

Quick Check

For each scenario, identify whether triadic, focal, or membership closure is at work.

1. Two students take the same elective and become friends.
2. A new graduate student joins the same lab as her advisor's other student, who introduces her.
3. A friend invites you to their book club, and you join.

Quick Check — Solution

1. — the elective is the shared focus that brings them into contact.
2. — the advisor is the shared friend that closes the triangle.
3. — your friend brings you into a focus you did not previously belong to.

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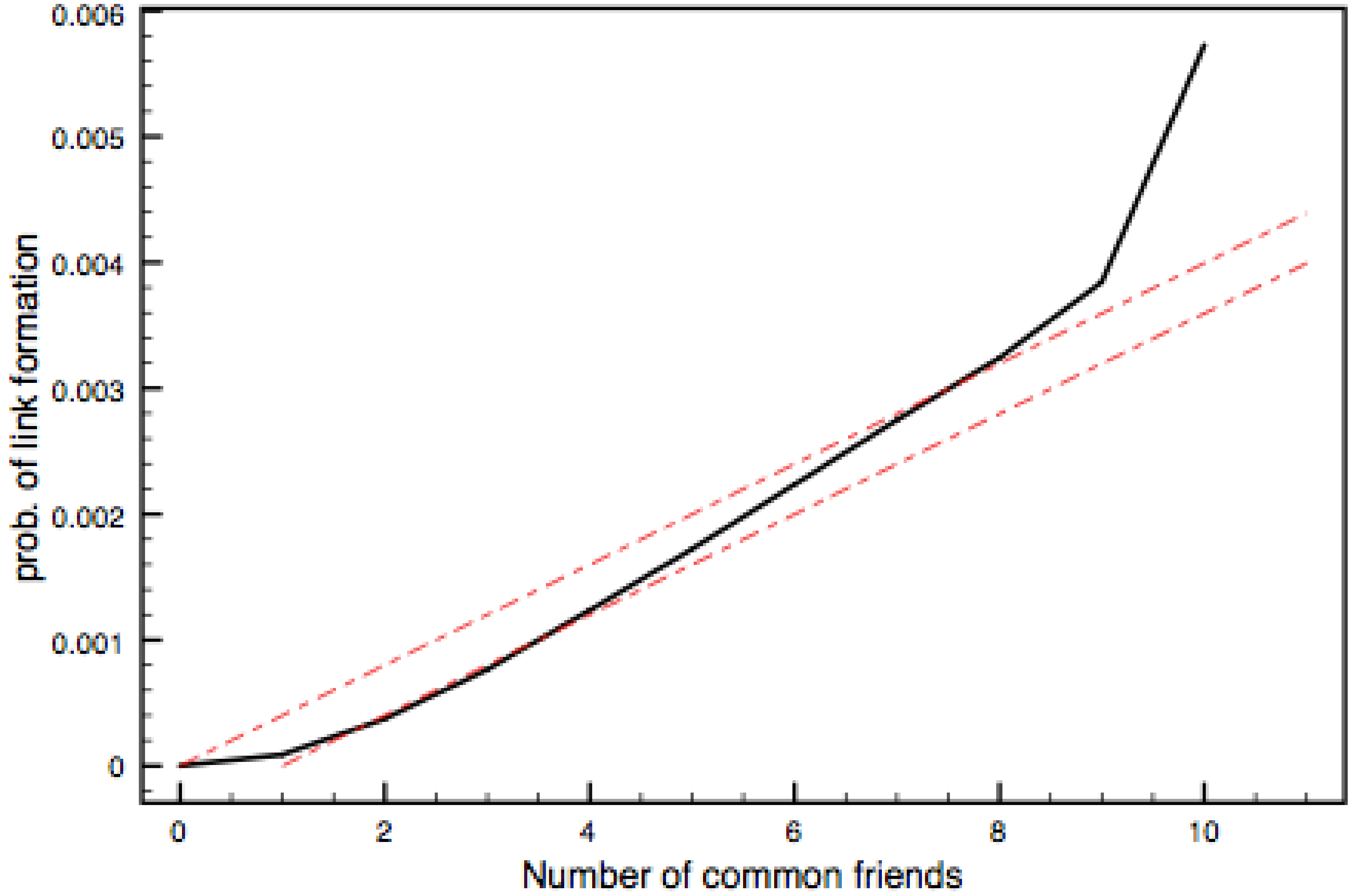
Takeaway

Affiliation networks make social context explicit. Focal closure and membership closure are distinct tie-formation mechanisms that can produce homophily-like patterns without any personal preference for similarity — making them a critical alternative explanation in any homophily analysis.

Online Link Formation

Guiding Question: What can large-scale online data teach us about how ties form over time?

Triadic Closure at Scale



Source: Easley & Kleinberg 2010

The probability of a new edge rises with the number of shared contacts — but with **diminishing returns**.

A Toy Model for the Shape

If each shared contact independently offers an introduction opportunity with probability q, then the probability of closure given k shared contacts is:

$$P(\text{new tie} \mid k \text{ shared}) = 1 - (1 - q)^k$$

This *saturating* curve rises steeply at first and flattens as k grows — matching the empirical shape qualitatively. But real curves often flatten *faster* than this model predicts, suggesting that shared contacts are not fully independent.

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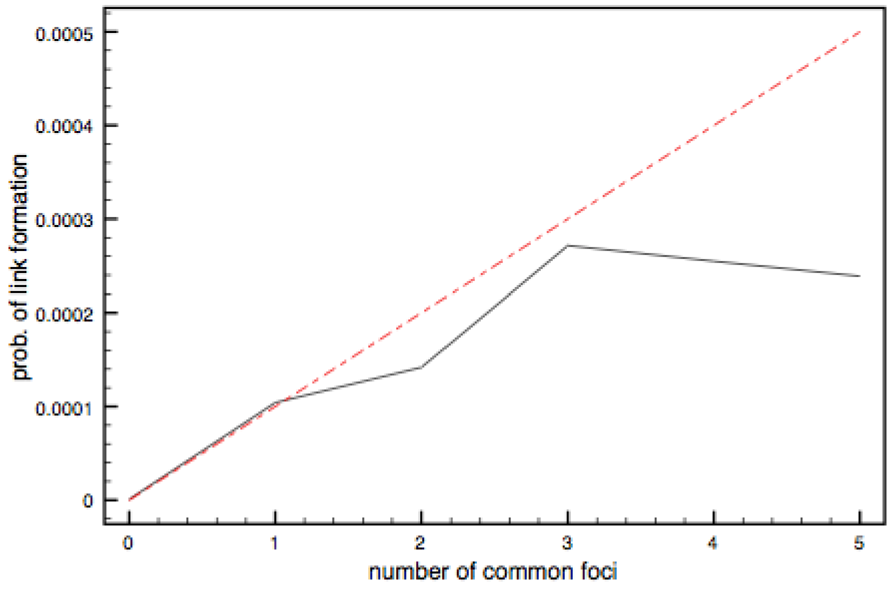
Example with $q = 0.20$:

Shared contacts k	1	2	3	5
$P(\text{new tie} \mid k)$	0.20	0.36	0.49	0.67

The first shared contact adds the most. Later shared contacts add less because many opportunities are redundant.

Focal and Membership Closure at Scale



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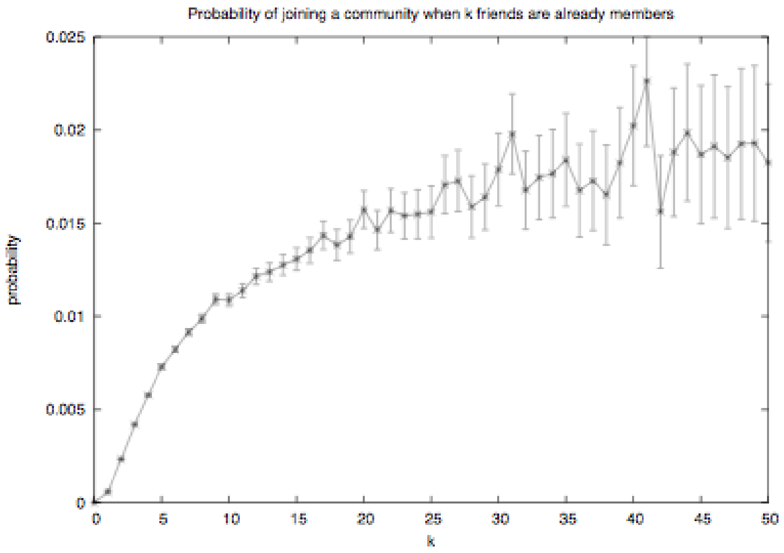

Source: Easley & Kleinberg 2010

Shared foci (left) and friend-mediated memberships (right) both raise the probability of new ties — again with diminishing returns. The shapes are qualitatively similar to triadic closure, reinforcing that all three closure mechanisms share a common structural logic.

Quick Check

A study finds that the probability of two coworkers becoming Facebook friends rises from 1% (no shared friends) to 20% (one shared friend) to 30% (two) to 33% (three or more).

1. Describe the shape of the curve.
2. Fit the independent-opportunity model: what value of q gives $P = 0.20$ at $k = 1$?
3. Does the model predict the observed $P = 0.30$ at $k = 2$? What does the mismatch tell you?

Quick Check — Solution

Steep initial rise from 1% to 20%, then much smaller gains (30%, 33%).

At $k = 1, P \approx 0.20 \Rightarrow q \approx 0.20$.

3. At $k = 2$ the independent model predicts $1 - (1 - 0.20)^2 = 0.36$, but we observe 0.30 — a shortfall. The shared friends are (typically embedded in the same group), so they provide less independent opportunity than the model assumes. The diminishing returns are steeper than independence predicts.

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3. At $k = 2$ the independent model predicts $1 - (1 - 0.20)^2 = 0.36$, but we observe 0.30 — a shortfall. The shared friends are *correlated* (typically embedded in the same group), so they provide less independent opportunity than the model assumes. The diminishing returns are steeper than independence predicts.

Takeaway

> Online data turns link formation into a measurable, time-stamped process. Empirical closure curves show saturation — the first shared contact matters most, and additional contacts are redundant rather than independent.

From Local Preferences to Global Segregation

> **Guiding Question:** Can weak local preferences generate strong global segregation?

Schelling's Threshold Model

Setup. Agents of two types are arranged on a grid (or network). Each agent i has a threshold $\tau \in [0, 1]$ — the minimum fraction of same-type neighbors it needs to be "satisfied".

Update rule.

1. Identify all *unsatisfied* agents (fewer than fraction τ of neighbors are same-type).
2. Each unsatisfied agent moves to a random vacant cell (or rewires a random tie in a network setting).
3. Repeat until no agent wants to move — or until a stable oscillation is reached.

Schelling's Threshold Model

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Schelling's insight (1971). Even τ as low as $1/3$ — "I just want a third of my neighbors to be like me" — produces **sharply segregated** global patterns.

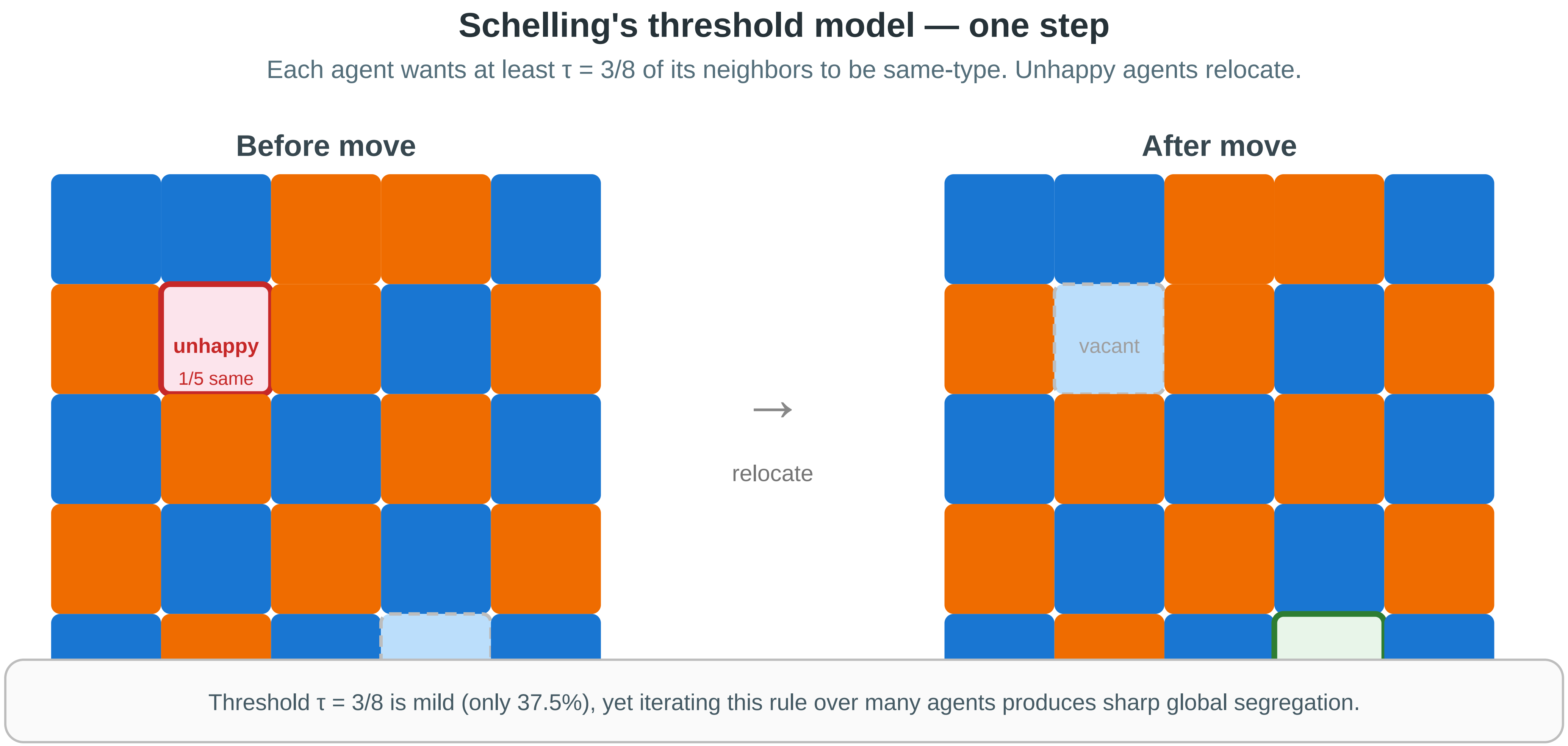
Schelling — A Local Rule

Let $\tau = 1/3$. An agent compares its local neighborhood to that threshold.

Same-type neighbors	Total neighbors	Same-type fraction	Status
2	8	0.25	Unsatisfied: move or rewire
3	8	0.375	Satisfied: stay
4	8	0.50	Satisfied: stay

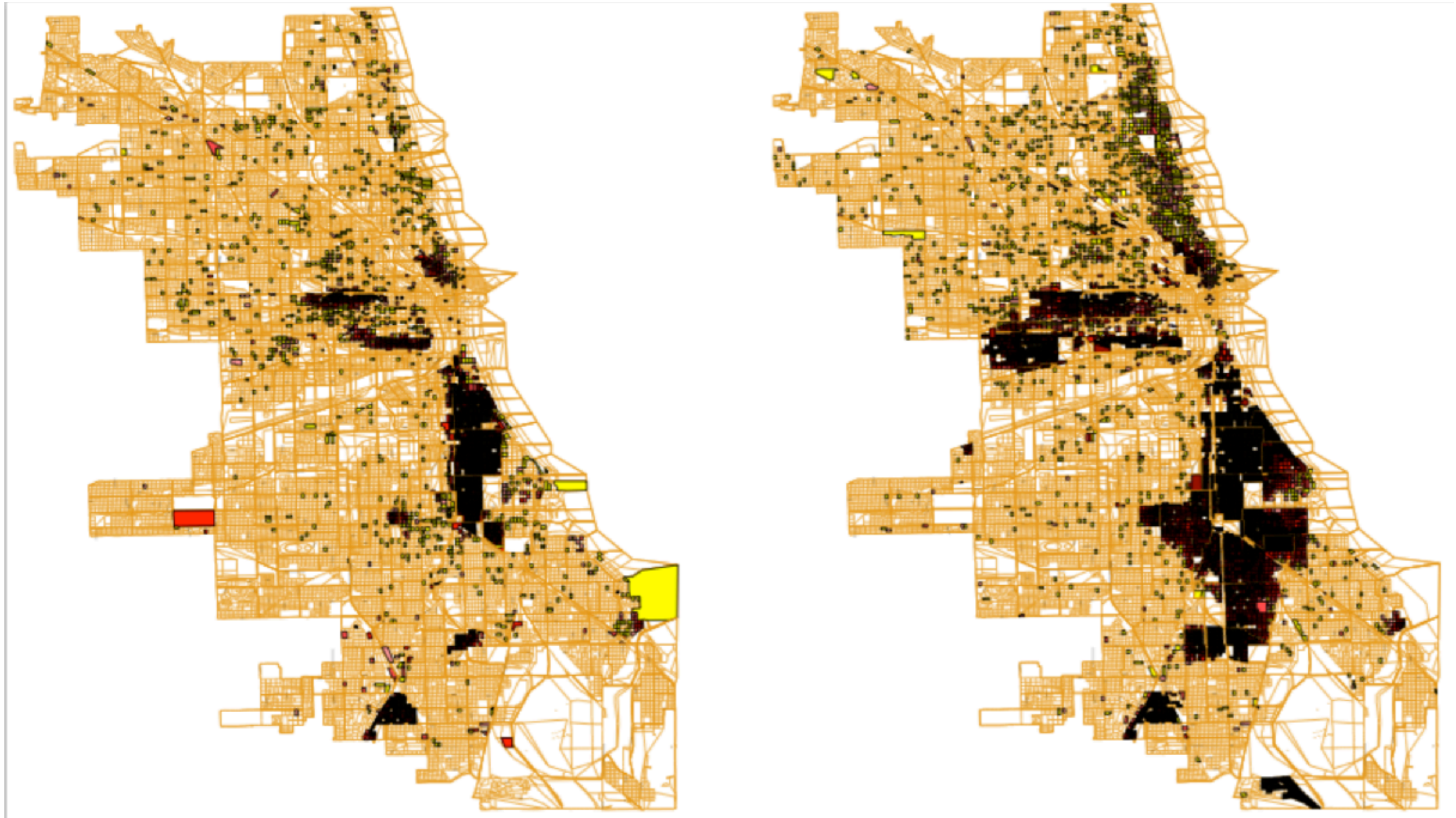
The rule is mild: the agent does **not** require a majority of similar neighbors. Strong segregation emerges because many mild local moves are repeated.

One Step of the Model



The blue agent at (1,1) has 1/5 same-type neighbors — below $\tau = 3/8$. It relocates to vacant cell (4,3), where 3/5 neighbors are same-type — satisfied. Repeat across all unsatisfied agents until equilibrium.

Empirical Motivation



(a) *Chicago, 1940* (b) *Chicago, 1960*

Source: Easley & Kleinberg 2010

Residential segregation in Chicago by race/ethnicity. The sharp boundaries between neighborhoods are consistent with Schelling dynamics: each household's mild preference for "some" same-type neighbors, iterated over decades of moves, produces stark global segregation.

From Grids to Networks

In a network setting, Schelling's model becomes a **rewiring** process:

- "neighbors" are graph-adjacent nodes
- "moving" becomes dropping a tie to a dissimilar contact and forming a new tie to a similar contact
- the threshold τ controls how much same-type concentration triggers rewiring

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- the threshold τ controls how much same-type concentration triggers rewiring

Implication for recommendation algorithms. If a platform suggests contacts who are *similar* to your current contacts, it acts as an automated Schelling rewirer — accelerating homophily-driven segregation even beyond what individual preferences would produce alone.

Quick Check

Translate Schelling's spatial model into a network setting.

1. What plays the role of "neighbor" in a network rather than a grid?
2. What plays the role of "moving"?
3. A social media platform recommends friends similar to your existing friends. In Schelling terms, what is the platform doing — and would you expect it to amplify or weaken homophily?

Quick Check — Solution

A graph-adjacent node — your immediate contacts.

Rewiring — dropping ties to dissimilar contacts and forming new ones with similar contacts.

3. The platform is an automated Schelling rewirer. It surfaces similar people, which is equivalent to offering each agent a move toward a higher same-type fraction. This homophily — the local rule ("recommend similar") is mild, but at machine speed it produces sharp global outcomes (filter bubbles, echo chambers).

Quick Check — Solution

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Takeaway

> Global polarization can emerge without anyone explicitly aiming for it. Mild local preferences plus iterated local updates produce sharp large-scale structure — and recommendation algorithms now run this process at scale.

Wrap-Up — The Three Gaps

Across Lectures 03–05, the course has exposed three kinds of gap between what we *want to know* about a network and what the data *actually shows*:

Lecture	Gap type	Theoretical ideal	Practical limitation	Strategy
L03	Computational	True tie labels (MaxSTC)	NP-hard	Polynomial proxies (clustering, overlap)
L04	Computational	Optimal partition ($\max Q$)	NP-hard	Heuristics (GN, Louvain, Leiden)
L05	Causal	Mechanism (selection vs. socialization)	Not identifiable from snapshots	Longitudinal data, affiliation models

Every concept in the three lectures occupies one cell of this table.

Looking Ahead: Lecture 06

If signs (positive and negative) — not just connection strengths — matter, how do whole social structures stabilize or destabilize? Structural balance theory turns the question of friend-and-enemy relations into a graph-theoretic claim with measurable predictions.

Reading

Chapter 4 of Easley & Kleinberg (2010) covers homophily, selection, and socialization. McPherson, Smith-Lovin & Cook (2001), *Birds of a Feather*, is the definitive review of homophily research. Christakis & Fowler (2007) is the Framingham obesity study; Lyons (2011) provides the methodological critique.

Image Credits & References

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